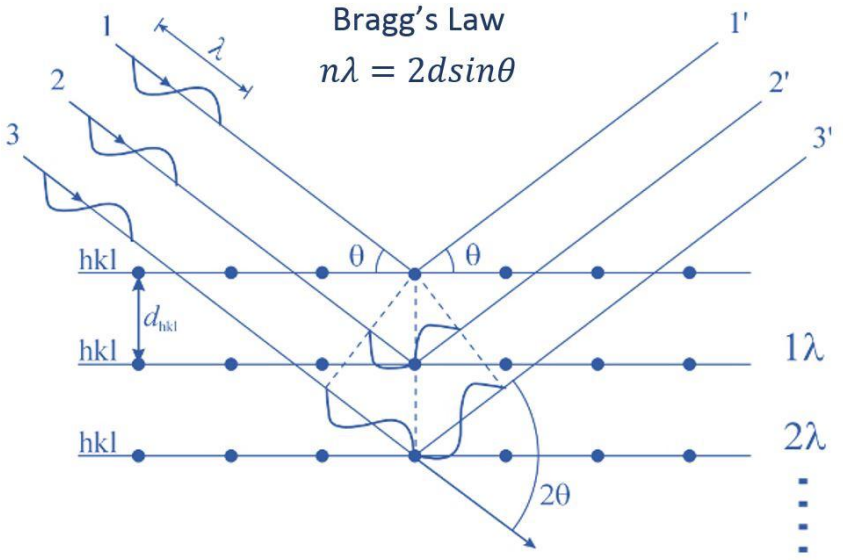
Fundamental Physics II
Optics - Chapter 2



For B1 Students
Academic year 2024-2025



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OVERVIEW

Chapter	Contents	Hours			Ref./Resources	Assignment(s)
		Lect.	Exr.	Prc.		
1	The nature and propagation of light	2.0	1.0			Exercises
2	Geometric optics	5.0	2.0			Exercises
3	Interference Midterm test (0.5h)	3.0	1.5			Exercises
4	Diffraction	3.0	1.5			Exercises
5	Photons: Light waves behaving as particles*					Exercises
		13	6			

Assessment/ Evaluation		
	Attendance/Attitude	**10%**
	Exercise(s)	0%
	Practicals	0%
	Mid-term test	**30 %**
	Final exam	**60 %**

Reference Literature:

[1] Young and Freedman – University Physics with Modern Physics 15th Edition (2020)

Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

2.2 Reflection at a spherical surface

2.3 Refraction at a spherical surface

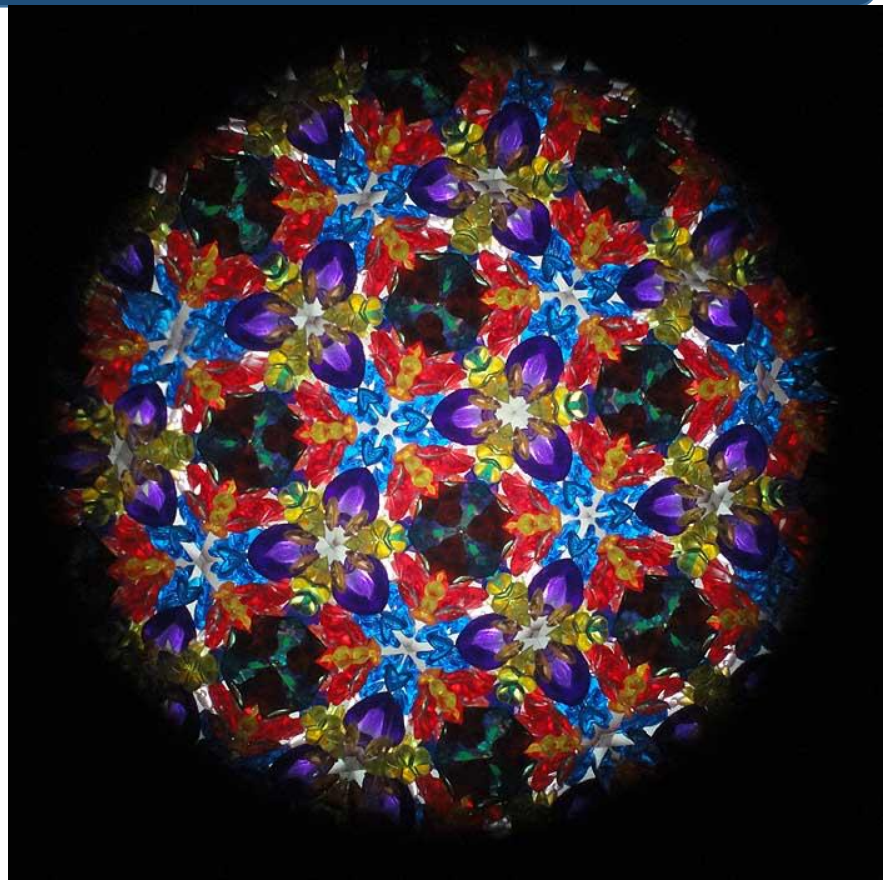
2.4 Thin lenses

2.5 Cameras

2.6 The eye

2.7 The magnifier

2.8 Microscopes and Telescopes



Kaleidoscope

List of Exercises:

In textbook “University Physics with Modern Physics” 15th Edition, Hugh D. Young, Roger A. Freedman, A. Lewis Ford (2020)

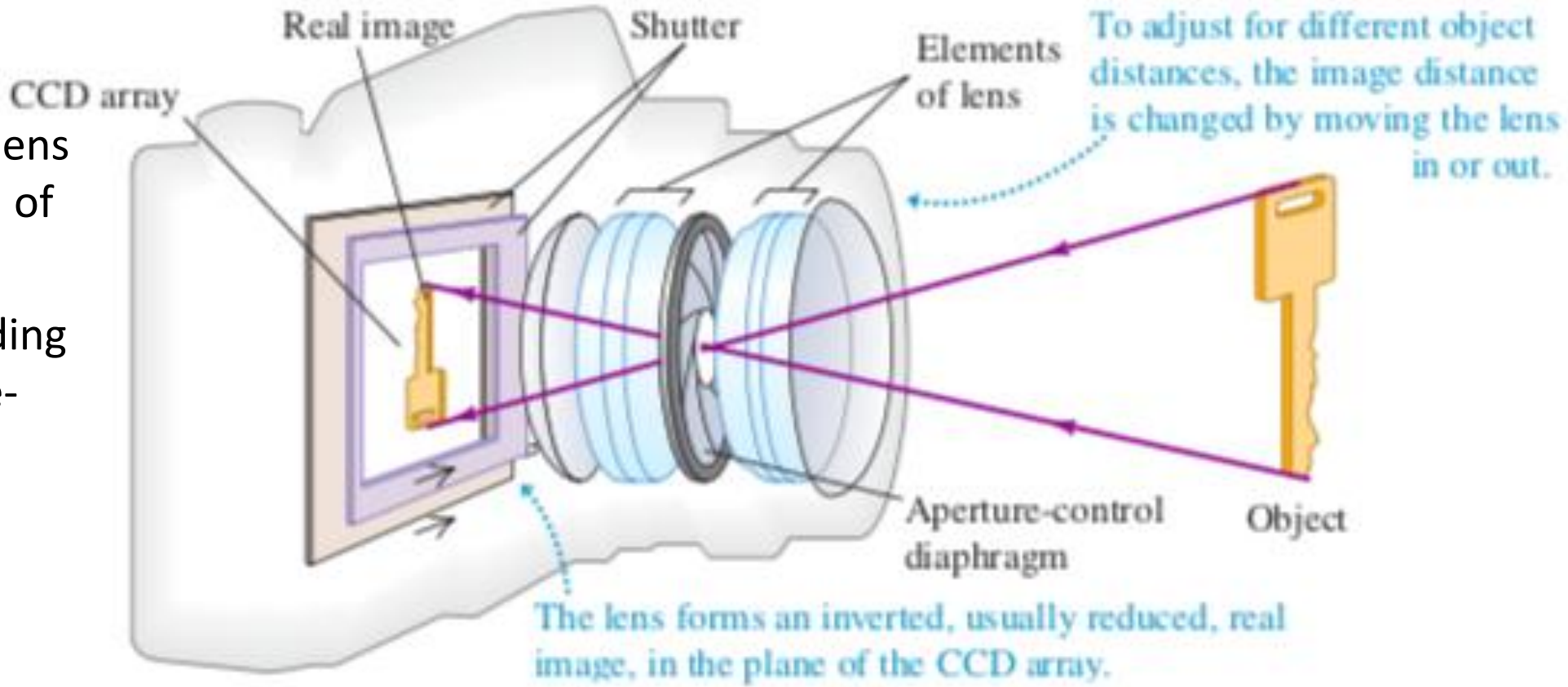
34. 3, 5, 8, 13, 19, 21, 26, 29, 31, 32, 40, 43, 45, 50, 52, 54, 58, 59, 62, 63, 66, 71

Chapter 2. Geometric optics (5.0h)

2.5 Cameras

"a room or enclosure" (Latin)

- A converging lens,
- A shutter to open the lens for a prescribed length of time,
- A light-sensitive recording medium (CCD - Charge-coupled device array; photographic film),

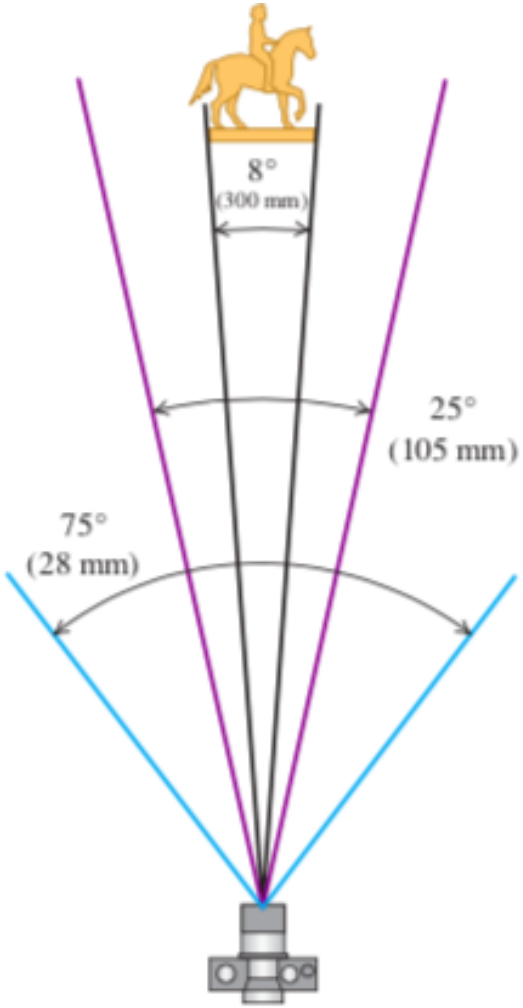


Chapter 2. Geometric optics (5.0h)

2.5 Cameras

2.5.1 Camera lenses: Focal length

The larger the value of f, the smaller the angle of view



(a) f = 28 mm



(b) f = 105 mm



(c) f = 300 mm



- A lens of *long focal length*, called a **telephoto** lens, gives a small angle of view and a large image of a distant object
- A lens of *short focal length* gives a small image and a wide angle of view and is called a **wide-angle** lens.

$$m = \frac{y'}{y} = -\frac{s'}{s}$$

Lateral magnification

Chapter 2. Geometric optics (5.0h)

2.5 Cameras

2.5.2 Camera lenses: *f*-Number

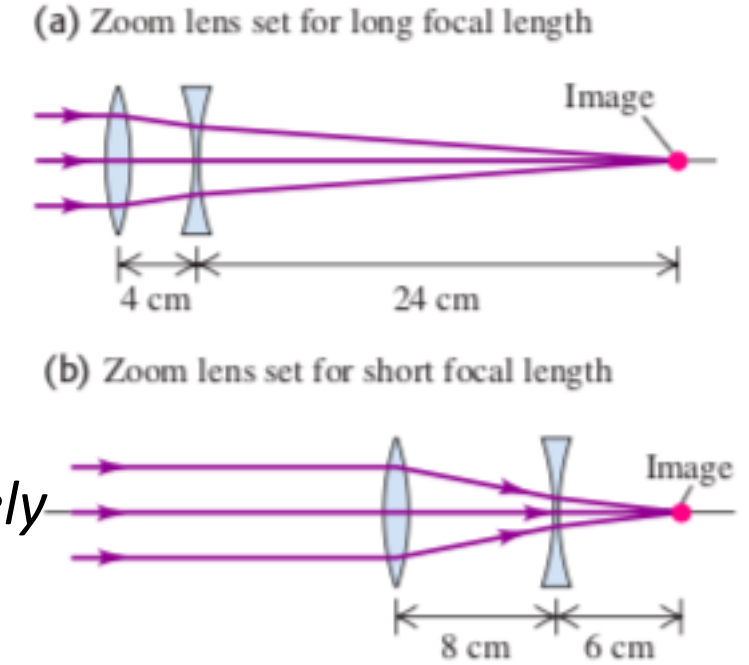
- The light-gathering capability of a lens is commonly expressed in terms of the ratio *f/D*: ***f*-number**

$$f-number = \frac{Focal\ length}{Aperture\ diameter} = \frac{f}{D}$$

f: focal length

D: the diameter of an adjustable lens aperture, or *diaphragm* (nearly circular hole)

- Exp: A lens with a focal length *f* = 50 mm and an aperture diameter *D* = 25 mm => *f*-number = 2; or "an aperture of *f*/2."
- The light intensity reaching the film is *inversely* proportional to the square of the *f*-number.





2.5.3 Zoom lenses and Projectors

- *Zoom lens,* which is not a single lens but a complex collection of several lens elements that give a continuously variable focal length, often over a range as great as 15 to 1
- *A projector* for viewing slides, digital images, or motion pictures operates very much like a camera in reverse

2.5 Cameras

Exp.2.8: Photographic exposures

A common telephoto lens for a 35-mm camera has a focal length of 200 mm; its *f*-stops range from *f*/2.8 to *f*/22.

(a) What is the corresponding range of aperture diameters?

(b) What is the corresponding range of image intensities on the film?

(a) The range of aperture diameters ranges

from $D = \dfrac{f}{f-number} = \dfrac{200}{22} = 9.1\ \text{mm}$

to $D = \dfrac{f}{f-number} = \dfrac{200}{2.8} = 71\ \text{mm}$

(b) The range image intensities on the film at *f*/2.8 and *f*/22 is

$$\left(\frac{71}{9.1}\right)^2 = 62$$

Chapter 2. Geometric optics (5.0h)

2.5 Cameras

Exp.2.9: Choosing a camera lens

The picture size on ordinary 35-mm camera film is 24 mm × 36 mm. Focal lengths of lenses available for 35-mm cameras typically include 28, 35, 50 (the "normal" lens), 85, 100, 135, 200, and 300 mm, among others. Which of these lenses should be used to photograph the following objects, assuming that the object is to fill most of the picture area?

(a) a building 240 m tall and 160 m wide at a distance of 600 m

(b) a mobile home 9.6 m in length at a distance of 40.0 m.

(a) A building 240 m tall and 160 m wide at a distance of 600 m

The lateral magnification: $m = \frac{y'}{y} = -\frac{24 \times 10^{-3}}{160} = -\frac{36 \times 10^{-3}}{240} = -1.5 \times 10^{-4}$

$s = 600\ \text{m} \gg f \Longrightarrow s' \approx f$ Because: $\left(\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}\right)$ $m = -\frac{s'}{s}$

$\Longrightarrow f = s' = -m \times s = 1.5 \times 10^{-4} \times 600\ (\text{m}) = 90\ (\text{mm})$ $\Longrightarrow$ Choose lens $f = 85\ \text{mm}$

(b) A mobile home 9.6 m in length at a distance of 40.0 m

The lateral magnification: $m = \frac{y'}{y} = -\frac{36 \times 10^{-3}}{9.6} = -3.75 \times 10^{-3}$

$\Longrightarrow f = s' = -m \times s = 3.75 \times 10^{-3} \times 40(\text{m}) = 150\ (\text{mm})$ $\Longrightarrow$ Choose lens $f = 135\ \text{mm}$

Chapter 2. Geometric optics (5.0h)

2.6 The Eye

- 2.5 cm diameter sphere

- *Cornea*: a tough, transparent membrane

- *Aqueous humor*: liquid region behind cornea

- *Crystalline lens*:

+ a capsule containing a fibrous jelly, hard at the center and progressively softer at the outer portions.
+ held in place by ligaments that attach it to the ciliary muscle, which encircles it.

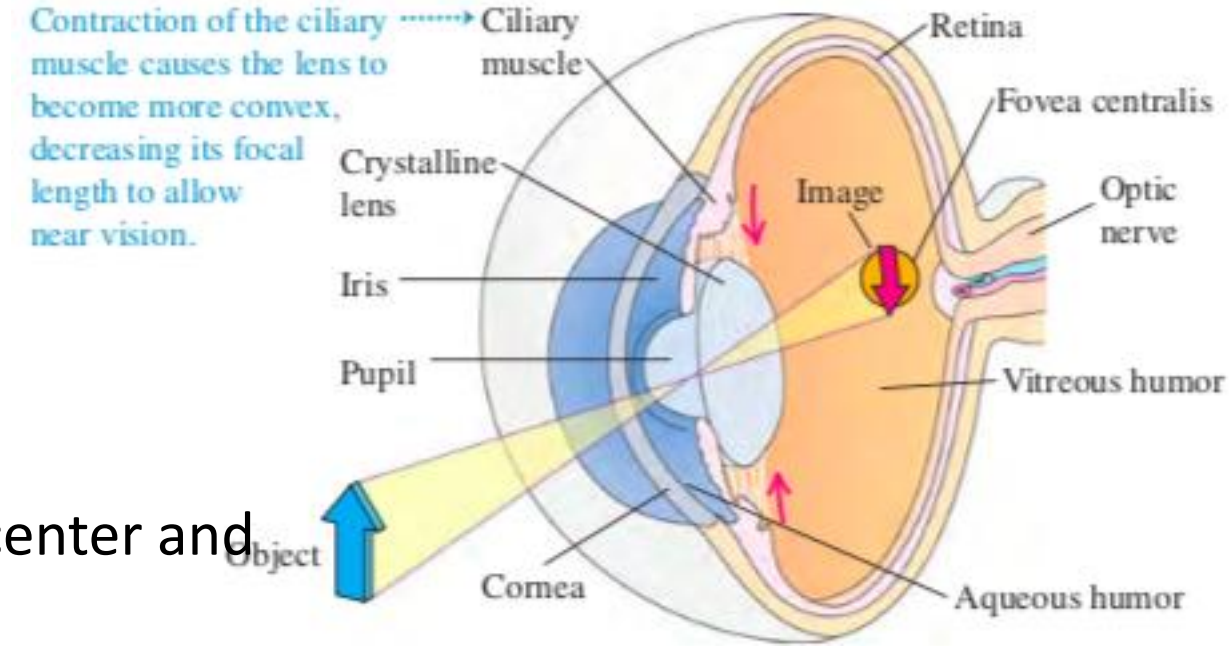


Diagram of the eye

- *Vitreous humor*: The aqueous humor and the vitreous humor : $n \sim 1.336$

+ the crystalline lens is not homogenous, $n \sim 1.437$
+ behind the lens, the eye is filled with a thin watery jelly.

- Most of the refraction of light entering the eye occurs at the outer surface of the cornea.

- *Real image:* refraction at the cornea and the surfaces of the lens

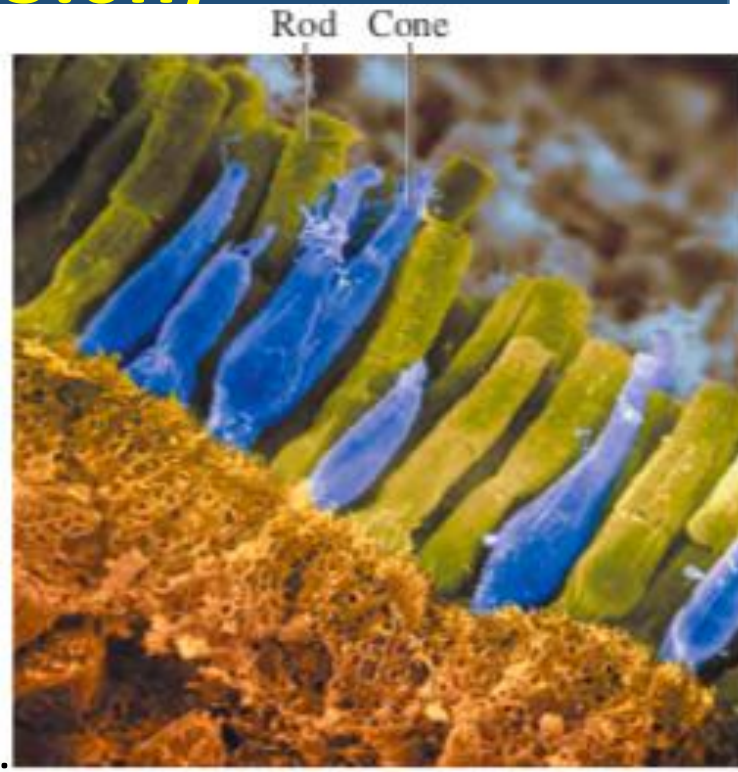
Chapter 2. Geometric optics (5.0h)

2.6 The Eye

- *Retina:* the image is formed on the light-sensitive, lining the rear inner surface of the eye ⇔ the film in a camera.
- The *rod - cones*: an array of miniature photocells.

=> sense the image and transmit it via the *optic nerve* to the brain.

- The *fovea centralis*:
  + about 0.25 mm in diameter
  + acute vision in it
- The *iris*:
  + contains an aperture with variable diameter called the *pupil*
  + *Pupil:* opens and closes to adapt to changing light intensity
  + the receptors of the retina also have intensity adaptation mechanisms.



Scanning electron micrograph showing retinal rods and cones in different colors

- *Accommodation*:

+ the eye adjusts to different object distances *s* by changing the length *f* of its lens; the lens-to-retina distance *s'* does not change (the image must be formed exactly at the location of the retina).

+ An object at infinity is sharply focused when the ciliary muscle is relaxed.

+ An object at closer, the tension in the ciliary muscle surrounding the lens ↑, the ciliary muscle ↓, focal length ↓

Chapter 2. Geometric optics (5.0h)

2.6 The Eye

- *Far/near point:*

the extremes of the range over which distinct is possible.

- The *far point* of a normal eye is at infinity.
- The *near point* of a normal eye is at 25 cm.
- The *near point* depends on the amount by which the ciliary muscle can increase the curvature of the crystalline lens.
- *Presbyopia*: the near point recedes with age because the crystalline grows throughout a person's life.

Table 34.1 Receding of Near Point with Age

"presbyopia"

Age (years)	Near Point (cm)
10	7
20	10
30	14
40	22
50	40
60	200

Chapter 2. Geometric optics (5.0h)

2.6 The Eye

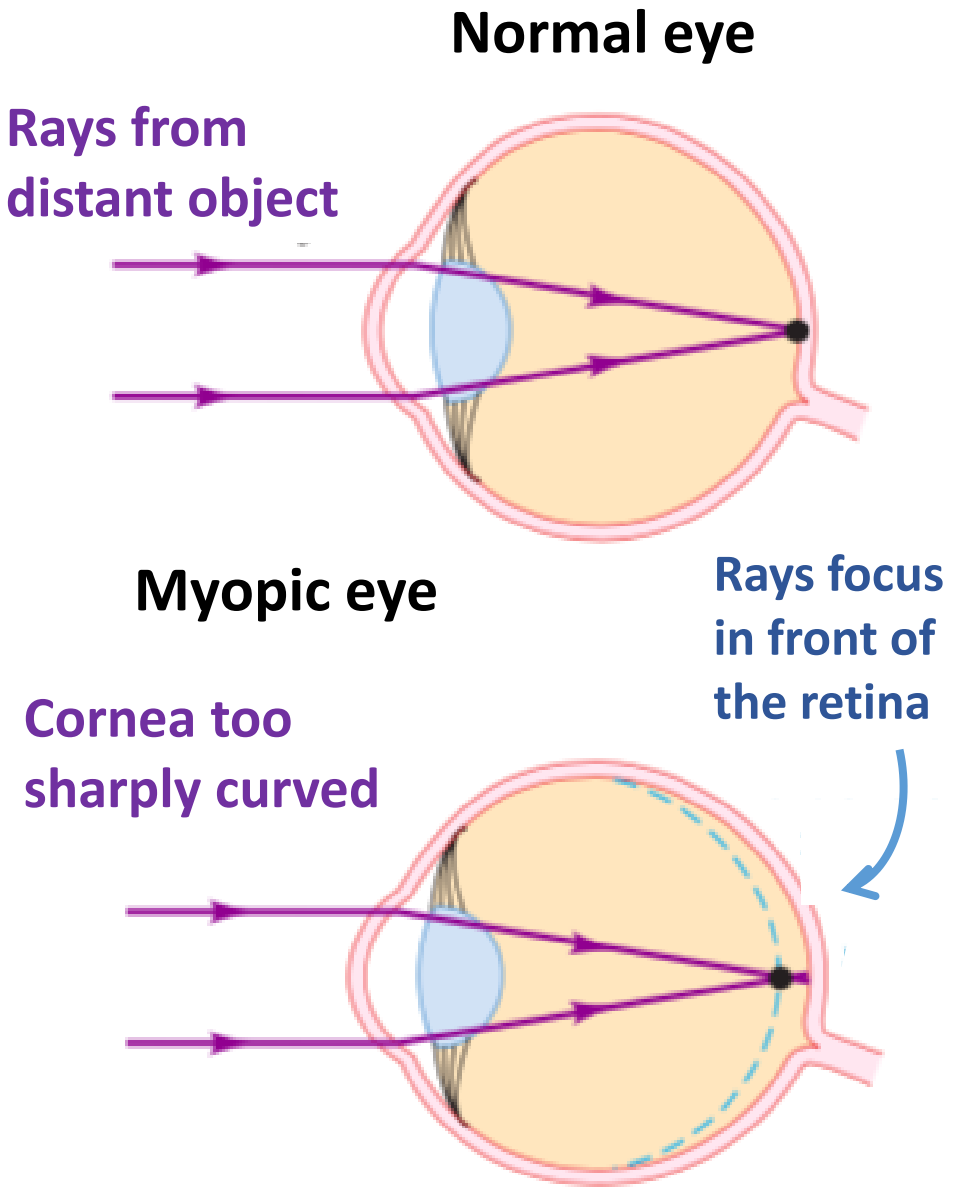
2.6.1 Defects of vision

Myopic (nearsighted) eye:

- The eyeball > the radius of curvature of the cornea

⇒ rays from an object at infinity are focused in front of the retina.

⇒ the most distant object for which an image can be formed on the retina is then nearer than infinity.



Chapter 2. Geometric optics (5.0h)

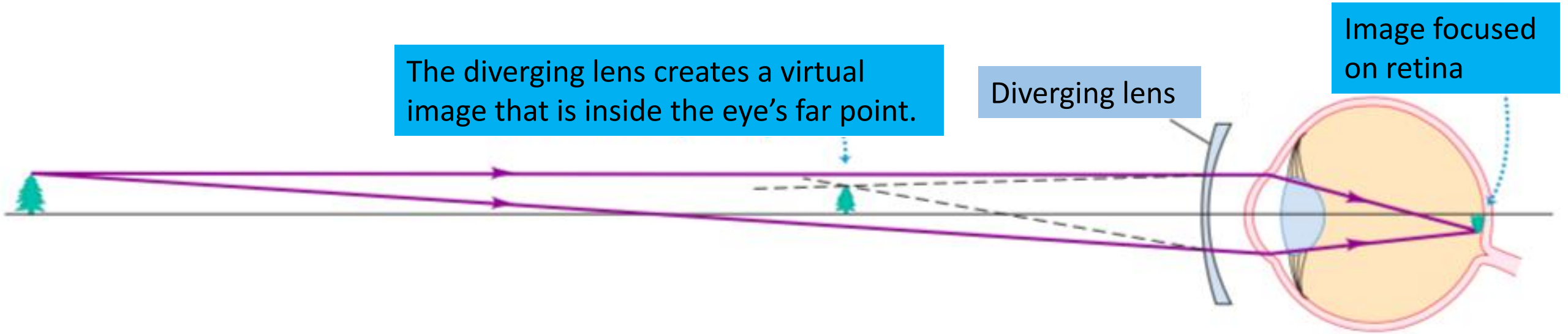
2.6 The Eye

2.6.1 Defects of vision

Myopic (nearsighted) eye:

- Correcting the **myopic eye** involves using a diverging (negative) lens to move the image closer to the eye than the actual object





Chapter 2. Geometric optics (5.0h)

2.6 The Eye

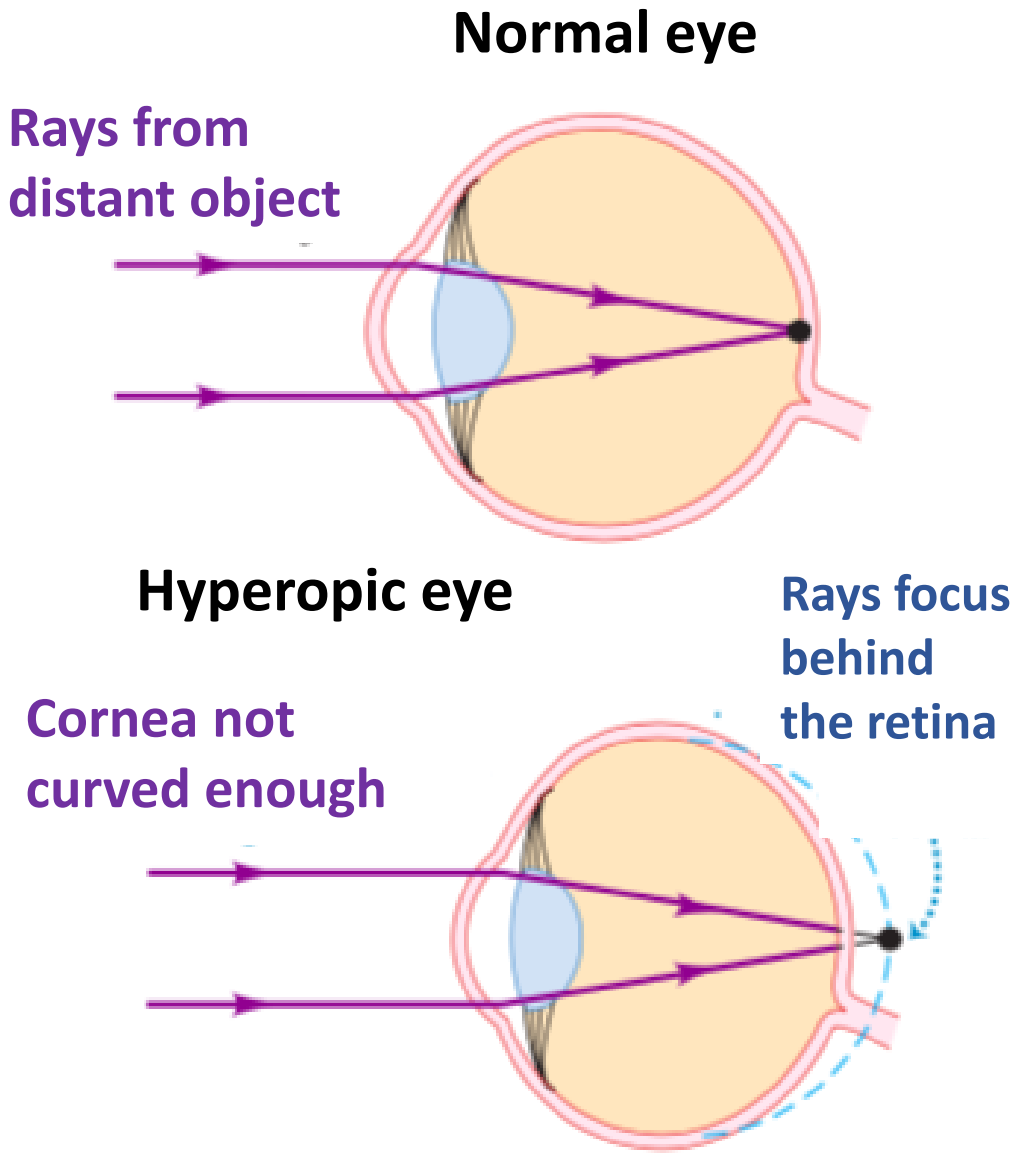
2.6.1 Defects of vision

Hyperopic (farsighted) eye:

- The eyeball < the radius of curvature of the cornea

  (The cornea is not curved enough)

⇒ rays from an object at infinity are focused **behind** the retina.



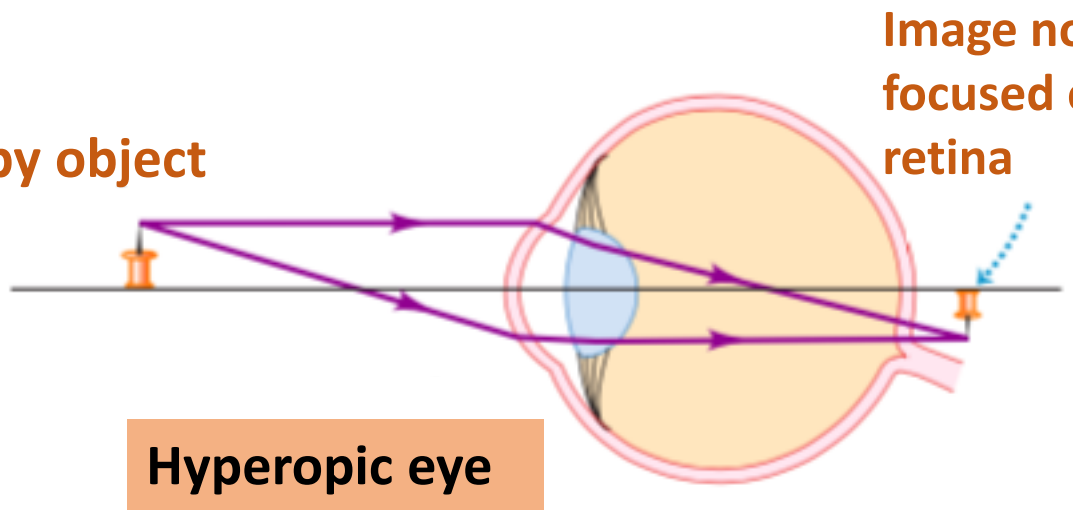
Chapter 2. Geometric optics (5.0h)

2.6 The Eye

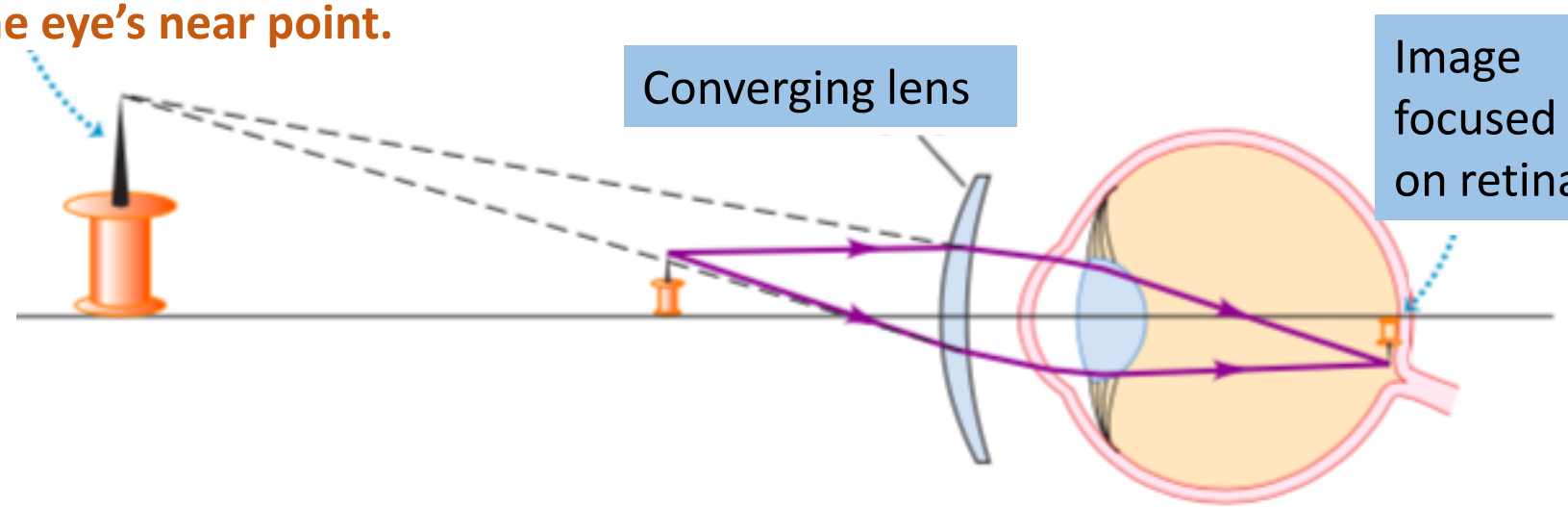
2.6.1 Defects of vision

Hyperopic (farsighted) eye:

- Correcting a **hyperopic eye** by using a converging (positive) lens that forms a virtual image of the object at or beyond the near point



A converging lens creates a virtual image at or beyond the eye's near point.



Chapter 2. Geometric optics (5.0h)

2.6 The Eye

2.6.1 Defects of vision

Astigmatism eye:

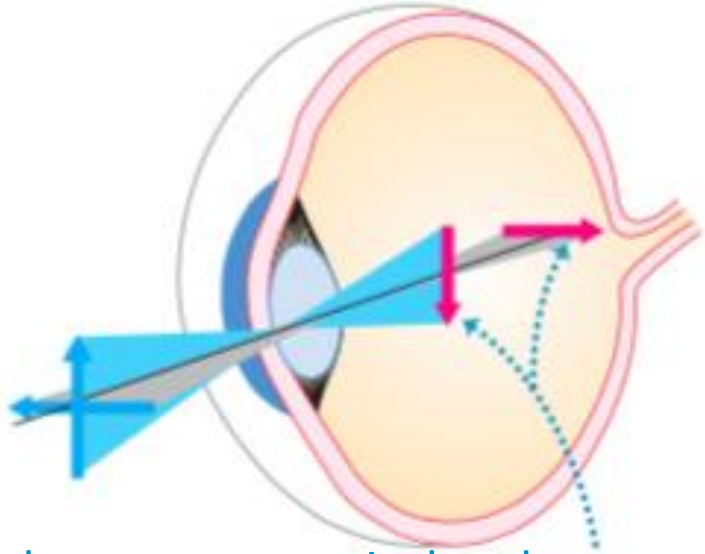
- The surface of the cornea is not spherical but rather more sharply curved in one plane than in another.

=> Horizontal lines may be imaged in a different plane from vertical lines

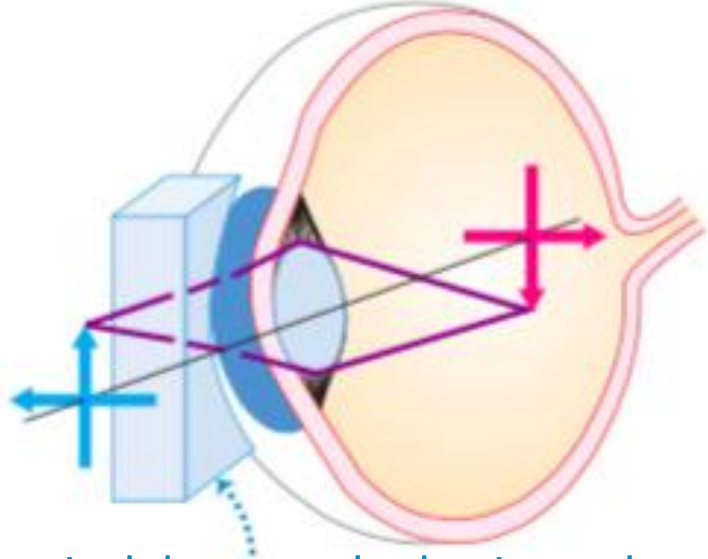
- **Astigmatism** can be corrected by use of a lens with a *cylindrical* surface
- When a cylindrical lens with its axis horizontal is placed before the eye:

+ the rays in a horizontal plane are unaffected,

+ the additional divergence of the rays in a vertical plane causes these to be sharply imaged on the retina



Shape of eyeball or lens causes vertical and horizontal elements to focus at different distances.



This cylindrical lens is curved in the vertical, but not the horizontal, direction; it changes the focal length of vertical elements.

Chapter 2. Geometric optics (5.0h)

2.6 The Eye

2.6.1 Defects of vision

- When the correction involves both astigmatism and myopia or hyperopia, there are three numbers for:

+ the spherical power,
+ the cylindrical power,
+ and an angle to describe the orientation of the cylinder axis.

- Lenses for vision correction are usually described in terms of the **power,**

  + defined as the reciprocal of the focal length expressed in meters.

  + The unit of power is the **diopter.**

$\Rightarrow$ A lens with f = 0.50 m has a power of 2.0 diopters,
f = -0.25 m corresponds to a power of - 4.0 diopters

Chapter 2. Geometric optics (5.0h)

2.6 The Eye

Exp.2.10 Correcting for farsightedness

The near point of a certain hyperopic eye is 100 cm in front of the eye. Find the focal length and power of the contact lens that will permit the wearer to see clearly an object that is 25 cm in front of the eye.

For the contact lens:

- The object distance s = 25 cm
- The image distance s’ = -100 cm

$$\frac{1}{s}+\frac{1}{s'}=\frac{1}{f}$$

Object s_1 = 25 cm | Contact Lens | Image1 (Object2) $s_1' = d - s_2$ | Eye s_2 = 100 cm | Image 2 at retina

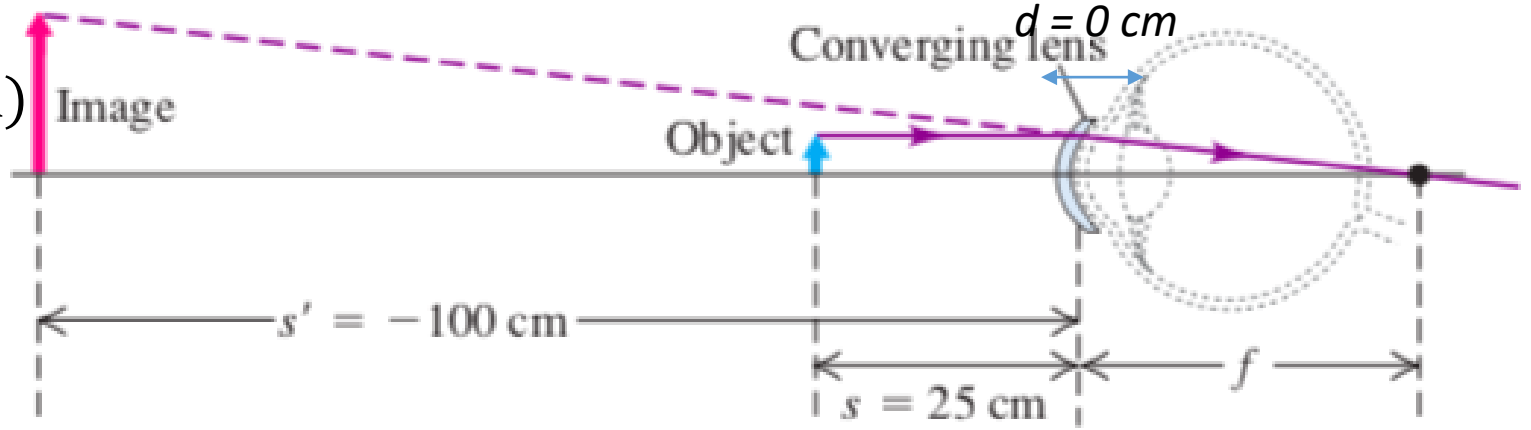
Contact lens: d = 0 cm (Distance between contact lens and the crystalline lens of the eye)

- The focal length f of the contact lens is

$$f=\frac{s\times s'}{s+s'}=\frac{25\times(-100)}{25-100}=\frac{100}{3}\,(\text{cm})$$

- The power of the contact lens is

$$\frac{1}{f}=\frac{1}{0.33}=3\ \text{diopters}$$



Chapter 2. Geometric optics (5.0h)

2.6 The Eye

Exp.2.11 Correcting for nearsightedness

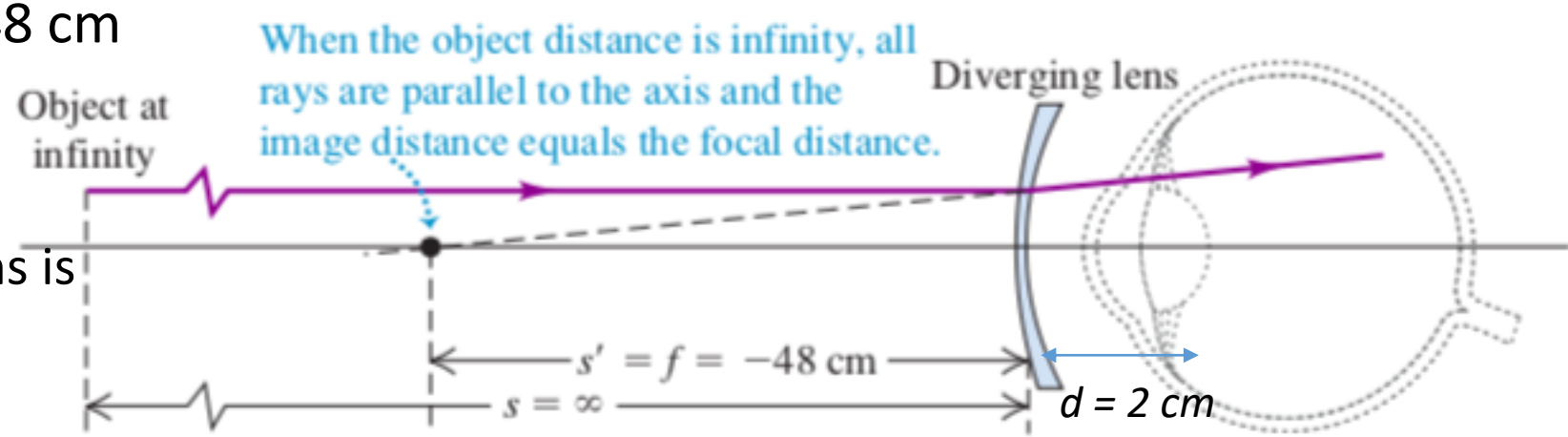
The far point of a certain myopic eye is 50 cm in front of the eye. Find the focal length and power of the eyeglass lens that will permit the wearer to see clearly an object at infinity. Assume that the lens is worn 2 cm in front of the eye.

For the eyeglass lens:

Object $s_1 = \infty$ Eyeglass Lens Image1 (Object2) $s_1' = d - s_2$ Eye $s_2 = 50$ cm Image 2 at retina

- The object distance $s = \infty$
- The image distance s' = 2-50 = -48 cm

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$



- The focal length f of the eyeglass lens is

$$f = s' = -48\ (\text{cm})$$

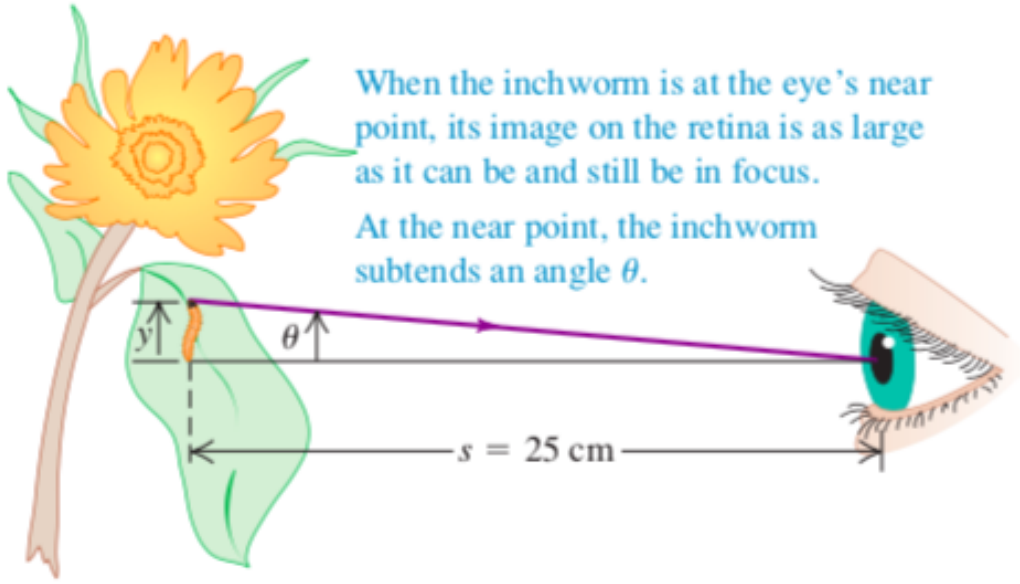
- The power of the eyeglass lens is

$$\frac{1}{f} = -\frac{1}{0.48} = -2.083\ \text{diopters}$$

Chapter 2. Geometric optics (5.0h)

2.7 The Magnifier

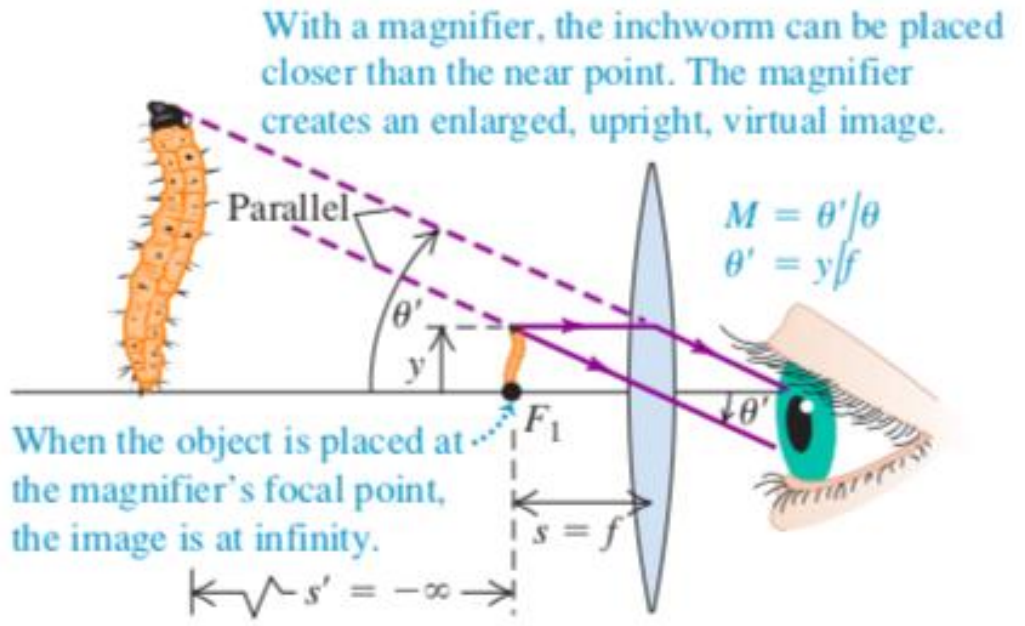
- **Angular size**: The size of the image on the retina depends on the angle θ subtended by the object at the eye.
  - The object is at the near point (25 cm) $\theta = \dfrac{y}{25\ \text{cm}}$
  - The object subtends an angle θ at the eye.



- A **magnifier** – A **magnifying glass** – A **simple magnifier:**

A converging lens can be used to form a virtual image that is larger and farther from the eye than the object itself.

- Forms an image at infinity
- The angle subtended at the magnifier is θ' $\theta' = \dfrac{y}{f}$



- **Angular magnification** *M*:

The ratio of the angle θ' (with the magnifier) to the angle θ (without the magnifier)

$$M = \frac{\theta'}{\theta} \quad \Rightarrow \quad M = \frac{\theta'}{\theta} = \frac{y/f}{y/25\text{cm}} = \frac{25\ \text{cm}}{f}$$

(Angular magnification for a simple magnifier)

Chapter 2. Geometric optics (5.0h)

2.8 Microscopes and Telescopes

2.8.1 Microscopes

- Components:

✓ **Objective lens** (focal length f_1)

=> forms a real image

✓ A **second lens – eyepiece** as a **magnifier** (focal length f_2)

=> makes an enlarged, virtual image

- Principle:

✓ Object *O* (**small**) is placed just beyond the 1st focal point F_1 of the **objective** ($s_1 \approx f_1$)

=> forms a real, enlarged image *I*. *I* lies just inside the 1st focal point F_2 of the eyepiece - *ocular*

✓ The eyepiece acts as a magnifier, forms a final virtual image *I'* of *I*.

✓ *I'* is between the near – far points of the eye.

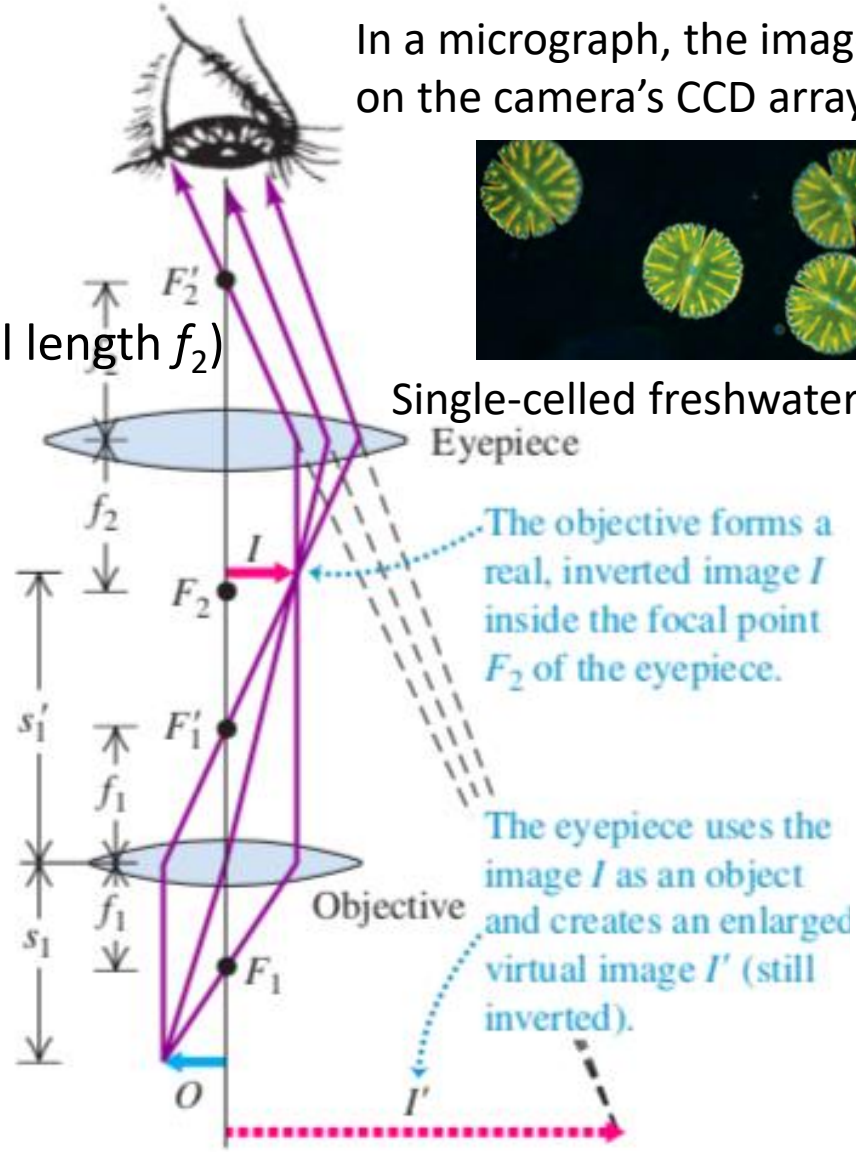
- Angular magnification *M*:

$$M = m_1 M_2 = \frac{25\text{cm} \times s'_1}{f_1 f_2} \qquad m_1 = -\frac{s'_1}{s_1} = -\frac{s'_1}{f_1}$$

m_1: lateral magnification of the objective lens

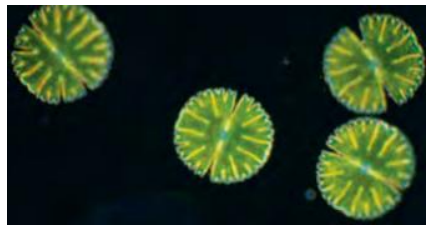
s_1': image distance for the objective lens

$$M_2 = \frac{25\text{cm}}{f_2}$$

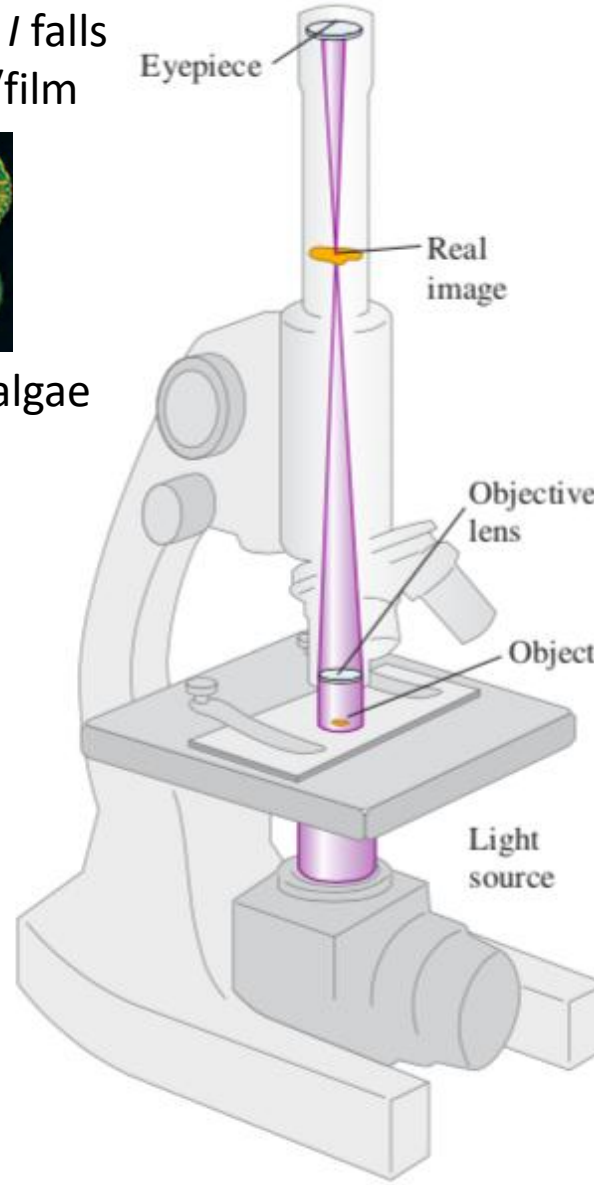


Mircoscope optics

In a micrograph, the image *I* falls on the camera's CCD array/film



Single-celled freshwater algae



Elements of a microscope

Chapter 2. Geometric optics (5.0h)

2.8 Microscopes and Telescopes

2.8.2 Telescopes

➢ Similar to the microscope

➢ View large objects at large distances

➢ Many telescopes use a curved mirror instead of lens as an objective

An astronomical telescope:

✓ The objective lens forms a real, reduced image *I* of the object

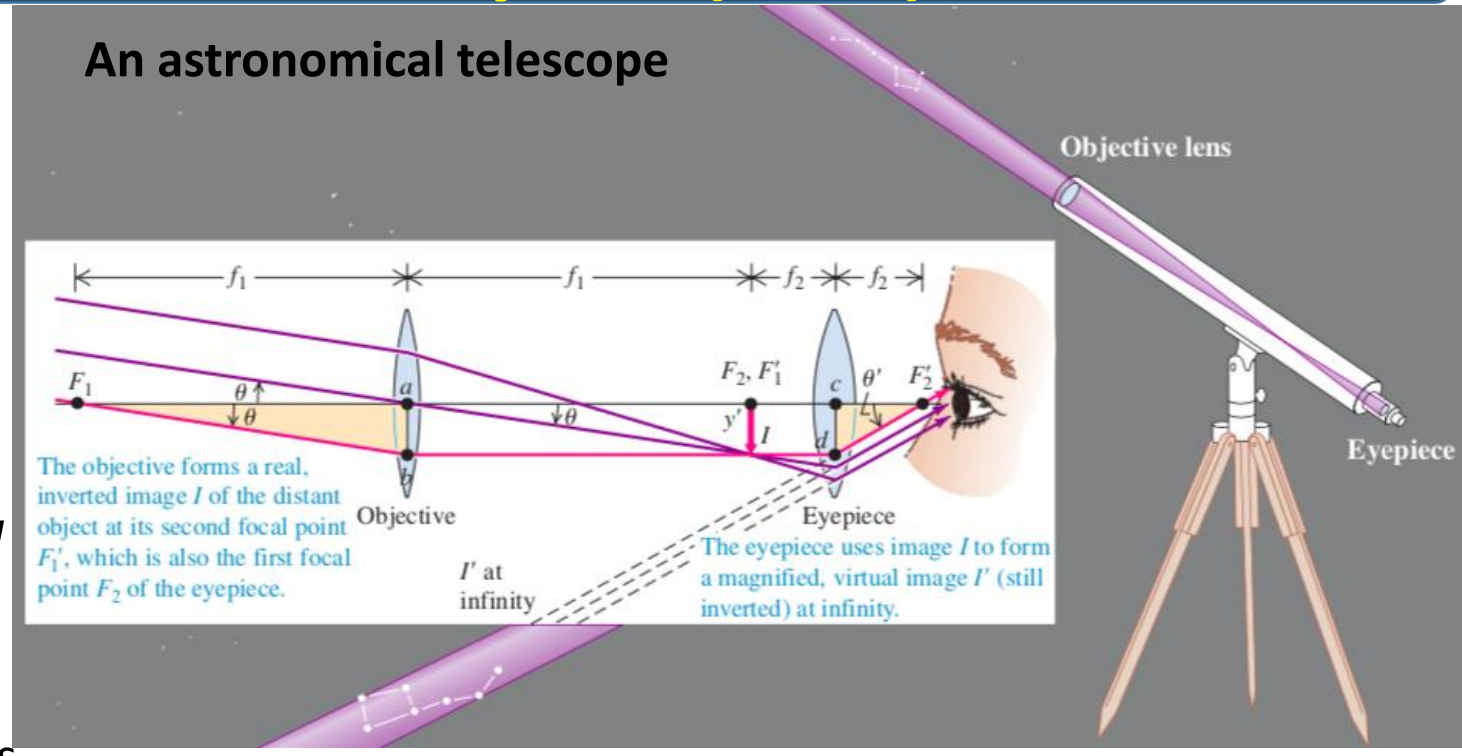
(Object at infinity => $I \approx F'_1$)

✓ *I* is the object for the eyepiece lens, which forms an enlarged, virtual image of *I*

(*I*' at infinity => $I \approx F_2$

✓ $f_1 + f_2$: the length of the telescope.

An astronomical telescope



Lens is used as an objective => "**refracting telescope**" or "**refractor**"

- When a telescope is used for photography the eyepiece is removed and a CCD array or photographic film is placed at the position of the real image formed by the objective.

- Angular magnification *M*:

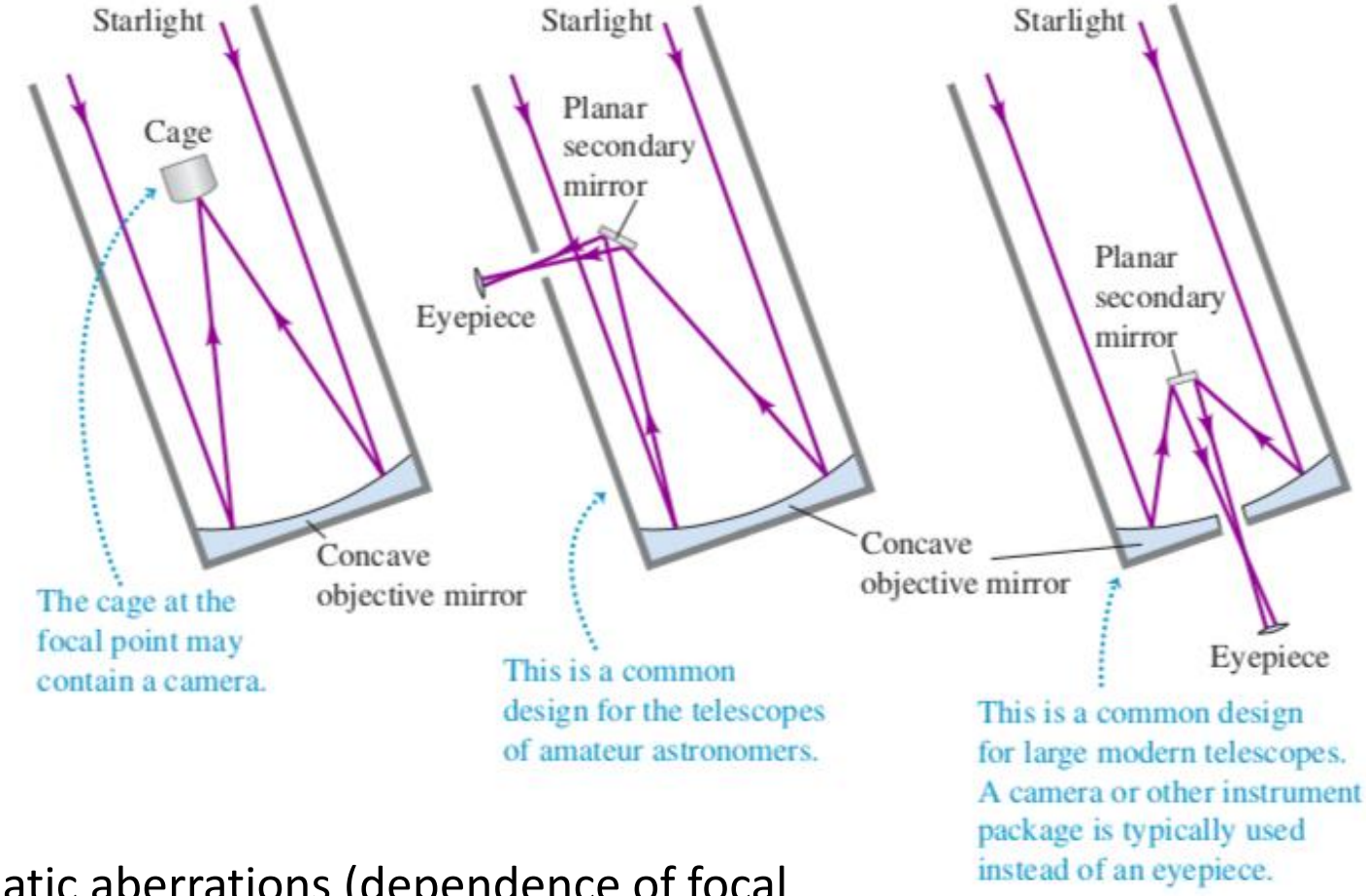
The ratio of the angle subtended at the eye by the final image *I*' to the angle subtended at the (unaided) eye by the object.

$$M = \frac{\theta'}{\theta} = -\frac{y'/f_2}{y'/f_1} = -\frac{f_1}{f_2}$$

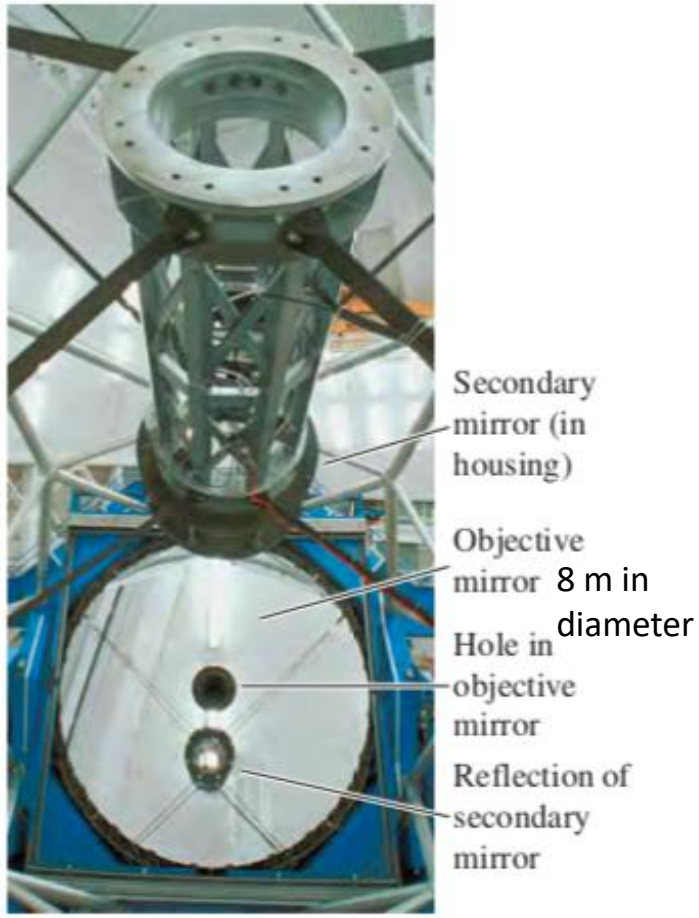
Chapter 2. Geometric optics (5.0h)

2.8 Microscopes and Telescopes

2.8.2 Telescopes



The interior of the Gemini North telescope



- Free of chromatic aberrations (dependence of focal length on wavelength),
- Spherical aberrations (associated with the paraxial approximation) are easier to correct than with a lens.

Mirrors is used as an objective => "**reflecting telescope**"

“When you really want something the whole universe conspires in helping you to achieve it”
(The Alchemist, Paulo Coelho)

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UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

GOOD LUCK and THANK YOU

Make it Simple but Significant!